QBS 103 Final Project - Submission 1

Ahhyun Lee

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Variables that I choose

- **Gene:** CXCR2
- **Continuous covariate:** maternal vitamin D at baseline (`mat_vitD`, ng/mL)
- **Categorical covariate 1:** pregnancy outcome (`preg_preeclampsia`: Normal vs Preeclampsia)
- **Categorical covariate 2:** trimester of blood draw (`collect_trimester`), derived from the continuous variable `sampleCollection_gestAge_wks` (1st trimester = before 14 weeks, 2nd trimester = 14 weeks or later)

Load Packages

```
# load ggplot2 for all plots
library(ggplot2)
```

Load the Data

All files are loaded in their original state, exactly as provided on Canvas.

```
# gene expression: rows are probe ids, columns are sample ids (gsm...)
expr <- read.csv(file = 'data/GSE85307_geneExpression.csv', row.names = 1)

# covariate data: one row per sample
covar <- read.csv(file = 'data/GSE85307_covar.csv')

# linker file between probe ids and gene names
geneNames <- read.csv(file = 'data/GSE85307_geneNames.csv')

# check the dimensions of each file
dim(expr)
```

```
## [1] 500 157
```

```
dim(covar)
```

```
## [1] 157  10
```

```r
dim(geneNames)
```

```
## [1] 750   2
```

```r
# take a peek at the top entries of each file
expr[1:3, 1:3]
```

```
##         GSM2264363 GSM2264379 GSM2264394
## 8089659   6.180152   6.105630    6.31849
## 8083594   4.803436   4.787737    4.80452
## 8035050   9.083982   8.902791    9.23949
```

```r
head(covar)
```

```
##   subjectID      geoID            mat_age_yrs                 preg_preeclampsia
## 1 6053887-M GSM2264346 age_years: 31.06639288 pregnancy_condition: Preeclampsia
## 2 3064425-M GSM2264347 age_years: 27.79466119       pregnancy_condition: Normal
## 3 6094898-M GSM2264348 age_years: 34.31895962       pregnancy_condition: Normal
## 4 7050617-M GSM2264349 age_years: 33.42642026       pregnancy_condition: Normal
## 5 7016015-M GSM2264350 age_years: 28.27926078 pregnancy_condition: Preeclampsia
## 6 7054360-M GSM2264351 age_years: 23.50171116       pregnancy_condition: Normal
##           delivery_gestAge_wks              sampleCollection_gestAge_wks
## 1 gestation_weeks: 37.42857143          blood_draw_at_gestation_week: 10
## 2 gestation_weeks: 39.42857143 blood_draw_at_gestation_week: 17.42857143
## 3 gestation_weeks: 38.85714286          blood_draw_at_gestation_week: 13
## 4 gestation_weeks: 39.14285714 blood_draw_at_gestation_week: 17.28571429
## 5 gestation_weeks: 40.14285714 blood_draw_at_gestation_week: 10.71428571
## 6 gestation_weeks: 41.14285714 blood_draw_at_gestation_week: 10.57142857
##                            batch_Microarray                mat_BMI
## 1 microarray_hybridization_date: 03/15/2013 bmi_kg/m2: 37.52407538
## 2 microarray_hybridization_date: 03/15/2013 bmi_kg/m2: 20.59144273
## 3 microarray_hybridization_date: 03/15/2013 bmi_kg/m2: 20.71118512
## 4 microarray_hybridization_date: 03/15/2013 bmi_kg/m2: 31.99358316
## 5 microarray_hybridization_date: 03/15/2013 bmi_kg/m2: 27.17734484
## 6 microarray_hybridization_date: 03/19/2013 bmi_kg/m2: 22.28684666
##                                        mat_vitD
## 1 vitamind_25(0h)_ng/ml_whole_blood_baseline: 7.94
## 2 vitamind_25(0h)_ng/ml_whole_blood_baseline: 40.3
## 3 vitamind_25(0h)_ng/ml_whole_blood_baseline: 40.8
## 4 vitamind_25(0h)_ng/ml_whole_blood_baseline: 52.2
## 5 vitamind_25(0h)_ng/ml_whole_blood_baseline: 22.8
## 6 vitamind_25(0h)_ng/ml_whole_blood_baseline: 44.6
##                         mat_asthma
## 1 asthma_0=healthy_1=asthmatic: 1
## 2 asthma_0=healthy_1=asthmatic: 1
## 3 asthma_0=healthy_1=asthmatic: 1
## 4 asthma_0=healthy_1=asthmatic: 0
## 5 asthma_0=healthy_1=asthmatic: 0
## 6 asthma_0=healthy_1=asthmatic: 0
```

```r
head(geneNames)
```

```
##   PROBEID      GENE
## 1 8128043      CNR1
## 2 7962489 PLEKHA8P1
## 3 7920271    S100A4
## 4 7900468      NFYC
## 5 8026496 LINC00661
## 6 7989725     PDCD7
```

Clean the Covariate Data

The first column name has a hidden character at the start of the file, and every covariate value starts with a label (e.g. `"age_years: 31.07"`), so both need to be fixed.

```r
# rename the first column (the hidden character at the start of the file changes its name)
colnames(covar)[1] <- 'subjectID'

# remove the label before ": " in every covariate column (columns 3 to 10)
# '.*: ' means "any characters followed by a colon and a space"
for (i in 3:ncol(covar)) {
  covar[, i] <- gsub(pattern = '.*: ', replacement = '', x = covar[, i])
}

# convert the continuous variables from character to numeric
covar$mat_age_yrs <- as.numeric(covar$mat_age_yrs)
covar$delivery_gestAge_wks <- as.numeric(covar$delivery_gestAge_wks)
covar$sampleCollection_gestAge_wks <- as.numeric(covar$sampleCollection_gestAge_wks)
covar$mat_BMI <- as.numeric(covar$mat_BMI)
covar$mat_vitD <- as.numeric(covar$mat_vitD)

# convert the categorical variables to factors with readable labels
covar$preg_preeclampsia <- factor(covar$preg_preeclampsia,
                                  levels = c('Normal', 'Preeclampsia'))
covar$mat_asthma <- factor(ifelse(covar$mat_asthma == 1, 'Asthma', 'Healthy'),
                           levels = c('Healthy', 'Asthma'))
covar$batch_Microarray <- factor(covar$batch_Microarray)

# derive a categorical variable from the continuous gestational age at blood draw
# the first trimester ends at 13 6/7 weeks, so < 14 weeks = 1st trimester
covar$collect_trimester <- factor(ifelse(covar$sampleCollection_gestAge_wks < 14,
                                         '1st trimester (<14 wks)',
                                         '2nd trimester (>=14 wks)'))

# check that the conversion worked (no missing values, sensible ranges)
summary(covar)
```

```
##      subjectID         geoID      mat_age_yrs       preg_preeclampsia
##  Length   :157   Length   :157   Min.   :18.43   Normal      :110
##  N.unique :157   N.unique :157   1st Qu.:22.15   Preeclampsia: 47
##  N.blank  :  0   N.blank  :  0   Median :26.57
##  Min.nchar:  9   Min.nchar: 10   Mean   :26.38
```

```
##  Max.nchar:  9   Max.nchar: 10   3rd Qu.:30.14
##                                  Max.   :38.76
##
##  delivery_gestAge_wks sampleCollection_gestAge_wks   batch_Microarray
##  Min.   :26.43        Min.   :10.00                 03/19/2013:45
##  1st Qu.:38.14        1st Qu.:12.14                 03/15/2013:44
##  Median :39.29        Median :13.43                 03/22/2013:29
##  Mean   :38.82        Mean   :14.13                 03/27/2013:26
##  3rd Qu.:40.00        3rd Qu.:16.43                 04/04/2013:10
##  Max.   :42.00        Max.   :18.86                 03/18/2013: 1
##                                                     (Other)   : 2
##     mat_BMI         mat_vitD       mat_asthma               collect_trimester
##  Min.   :17.95   Min.   : 5.11   Healthy:94   1st trimester (<14 wks) :87
##  1st Qu.:22.90   1st Qu.:13.40   Asthma :63   2nd trimester (>=14 wks):70
##  Median :27.33   Median :20.40
##  Mean   :28.94   Mean   :22.98
##  3rd Qu.:33.27   3rd Qu.:29.50
##  Max.   :61.95   Max.   :80.80
##
```

```
table(covar$collect_trimester)
```

```
##
##  1st trimester (<14 wks) 2nd trimester (>=14 wks)
##                       87                       70
```

Match Probe IDs to Gene Names

```
# check that every probe in the expression data is in the linker file
table(rownames(expr) %in% geneNames$PROBEID)
```

```
##
## TRUE
##  500
```

```
# keep only the probes that are in the expression data
geneNames <- geneNames[geneNames$PROBEID %in% rownames(expr), ]

# put both in the same order by probe id
expr <- expr[order(as.numeric(rownames(expr))), ]
geneNames <- geneNames[order(geneNames$PROBEID), ]

# check that the probe ids are now in the same order (should all be true)
table(rownames(expr) == geneNames$PROBEID)
```

```
##
## TRUE
##  500
```

```r
# replace the probe ids with gene names
rownames(expr) <- geneNames$GENE
expr[1:3, 1:3]
```

```
##         GSM2264363 GSM2264379 GSM2264394
## PHF13     7.197650   7.298383   7.147849
## TNFRSF8   8.094866   8.049821   7.929165
## CTRC      6.504210   6.674909   6.446666
```

Match Gene Expression to Covariates

The columns of the expression file are **not** in the same order as the rows of the covariate file, so the samples need to be put in the same order before combining.

```r
# check that the same samples are in both files
table(colnames(expr) %in% covar$geoID)
```

```
##
## TRUE
##  157
```

```r
table(covar$geoID %in% colnames(expr))
```

```
##
## TRUE
##  157
```

```r
# check whether they are already in the same order (they are not)
table(colnames(expr) == covar$geoID)
```

```
##
## FALSE
##   157
```

```r
# flip the expression data so that rows are samples and columns are genes
exprT <- as.data.frame(t(expr))

# put both data frames in the same order by sample id
exprT <- exprT[order(rownames(exprT)), ]
covar <- covar[order(covar$geoID), ]

# check that the sample ids are now in the same order (should all be true)
table(rownames(exprT) == covar$geoID)
```

```
##
## TRUE
##  157
```

```r
# combine the covariates and gene expression into one data frame
dat <- data.frame(covar, exprT)

# double check one sample: the value in the combined data should match the original file
dat$CXCR2[dat$geoID == 'GSM2264363'] == expr['CXCR2', 'GSM2264363']
```

```
## [1] TRUE
```

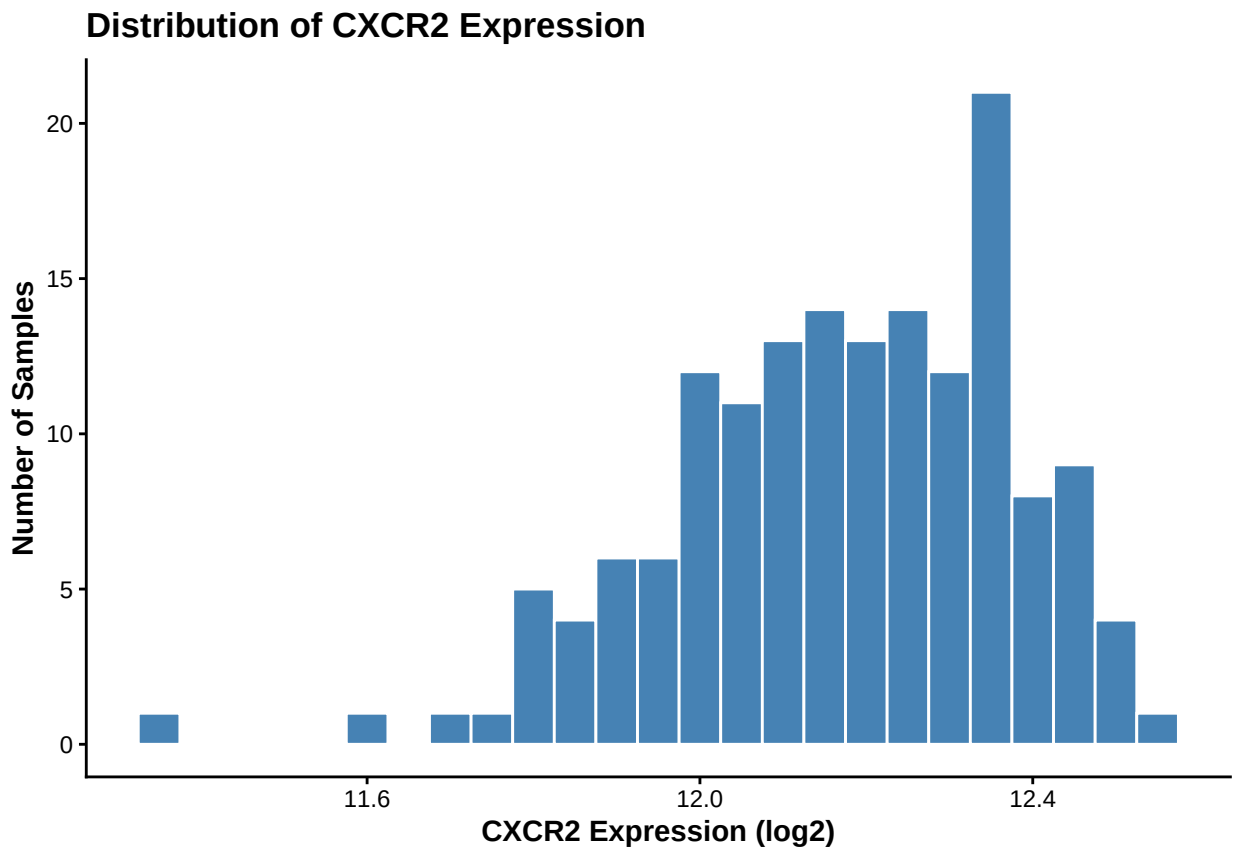
```r
# final sample size
dim(dat)
```

```
## [1] 157 511
```

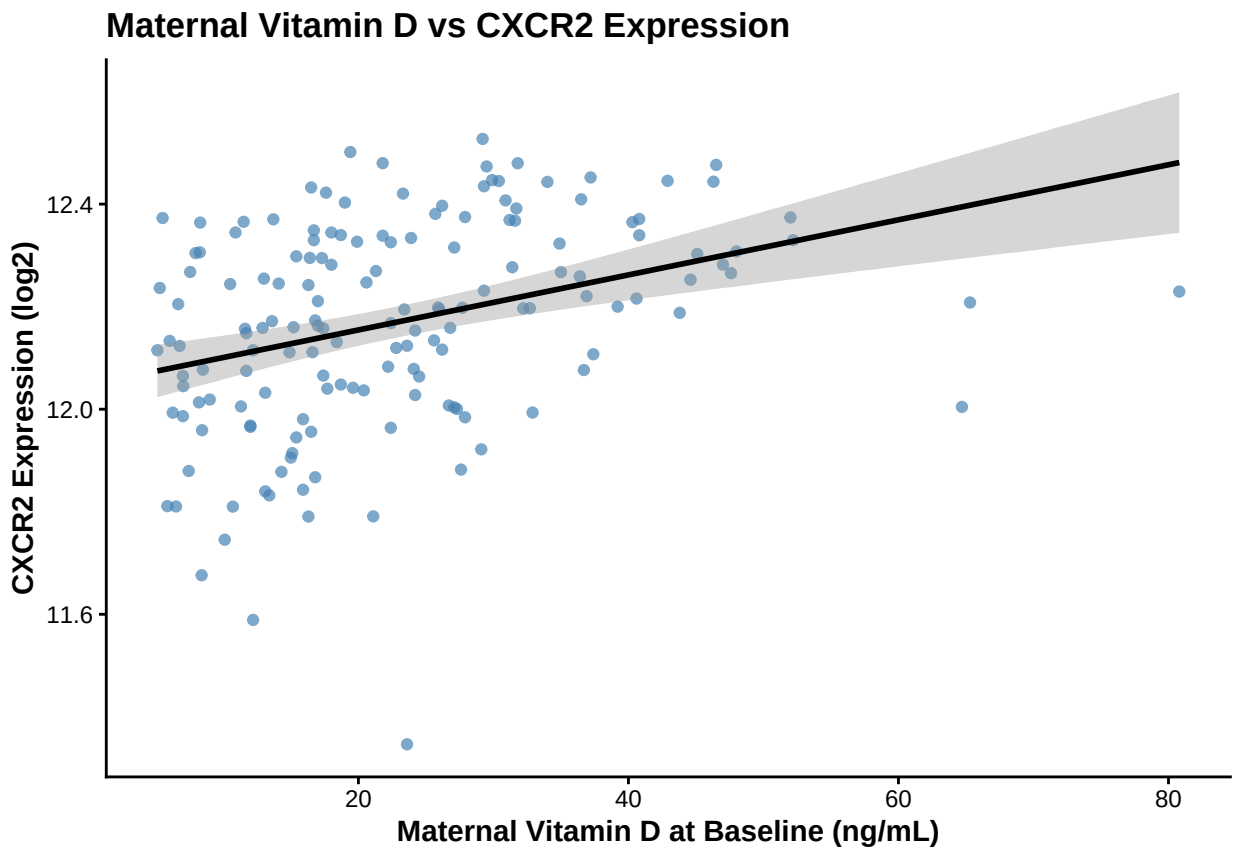
All 157 samples were matched, so no samples were removed.

Plots

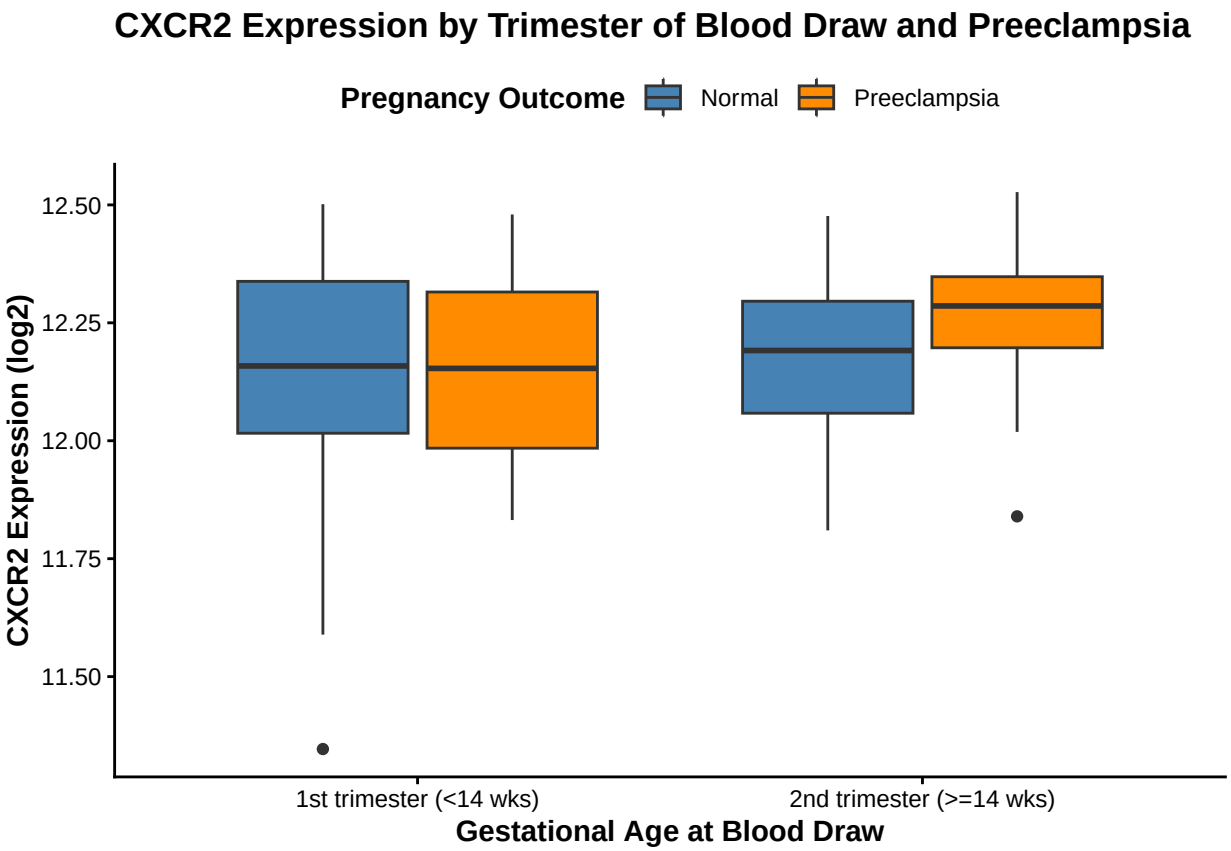
```r
# histogram of cxcr2 expression
ggplot(dat, aes(x = CXCR2)) +
  geom_histogram(binwidth = 0.05, fill = 'steelblue', color = 'white') +
  labs(title = 'Distribution of CXCR2 Expression',
       x = 'CXCR2 Expression (log2)',
       y = 'Number of Samples') +
  theme_classic() +
  theme(plot.title = element_text(face = 'bold'),
        axis.title = element_text(face = 'bold'))
```




```
# scatterplot of maternal vitamin d vs cxcr2 expression with a linear trend line
ggplot(dat, aes(x = mat_vitD, y = CXCR2)) +
  geom_point(color = 'steelblue', alpha = 0.7) +
  geom_smooth(method = 'lm', color = 'black') +
  labs(title = 'Maternal Vitamin D vs CXCR2 Expression',
       x = 'Maternal Vitamin D at Baseline (ng/mL)',
       y = 'CXCR2 Expression (log2)') +
  theme_classic() +
  theme(plot.title = element_text(face = 'bold'),
        axis.title = element_text(face = 'bold'))
```




```r
# boxplot of cxcr2 expression by trimester of blood draw and pregnancy outcome
ggplot(dat, aes(x = collect_trimester, y = CXCR2, fill = preg_preeclampsia)) +
  geom_boxplot() +
  scale_fill_manual(values = c('steelblue', 'darkorange')) +
  labs(title = 'CXCR2 Expression by Trimester of Blood Draw and Preeclampsia',
       x = 'Gestational Age at Blood Draw',
       y = 'CXCR2 Expression (log2)',
       fill = 'Pregnancy Outcome') +
  theme_classic() +
  theme(plot.title = element_text(face = 'bold'),
        axis.title = element_text(face = 'bold'),
        legend.title = element_text(face = 'bold'),
        legend.position = 'top')
```




```
# number of samples in each box
table(dat$collect_trimester, dat$preg_preeclampsia)
```

```
##
##                             Normal Preeclampsia
##   1st trimester (<14 wks)       58           29
##   2nd trimester (>=14 wks)      52           18
```